The set of cuts has the completeness property

The Open Logic Project contributors

openlogic:sets-functions-relations/arithmetization:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author's own theorem environment realcompleteness (source lines 944-946, the \ollabel on line 944); the proof environment begins two lines below it and is its own proof, closing at source line 968.

Editorial note. The proof contains a commented-out line of the source (a disabled display of the union); the retained span reproduces the source verbatim, comments included, and nothing was removed or added.

Editorial note. Neither the statement nor the proof contains an \olref, an \Olref or a \cref, so no source pointer and no bibliography entry are needed; the cut and order notations are declared in the retained preamble.

Editorial note. The retained preamble is the minimal working preamble shipped with the frozen OLP unit (OLP's own style stack needs prooftree, ntheorem, extarrows, mathdots, isomath and subfiles, which the project toolchain does not have). Author text is untouched; only the packaging is the intake line's.

Editorial note. Section and theorem counters are reset to zero so the retained theorem prints with the numbering of the retained front matter rather than continuing the frozen file's count.

Editorial scope. One original result of the Open Logic Project's arithmetization unit, counted once: the cuts have the completeness property.

Allowed standard prerequisites (editorial).

- Cut (Dedekind cut). A cut is a non-empty proper subset of the rationals that is downward closed under the order and has no greatest element.. The first half of the proof verifies the three clauses of this definition for the union lambda (source lines 952-965).
- Upper bound of a set of cuts. A cut beta is an upper bound of a set S of cuts when every member of S is contained in beta.. The hypothesis that S has an upper bound is used twice: to obtain a rational outside lambda (source lines 956-958) and, in the form of a general beta, to show that lambda is the least upper bound (source lines 966-967).

Theorem 0.1. *The set of cuts has the Completeness Property.*

Proof. Let S be any non-empty set of cuts with an upper bound. Let $\lambda = \bigcup S$. We first claim that λ is a cut:

1. Since S has an upper bound, at least one cut is in S, so $\emptyset \neq \lambda$. Since S is a set of cuts, $\lambda \subseteq \mathbb{Q}$. Since S has an upper bound, some $p \in \mathbb{Q}$ is absent from every cut $\alpha \in S$. So $p \notin \lambda$, and hence $\lambda \subsetneq \mathbb{Q}$.

2. Suppose $p < q \in \lambda$. So there is some $\alpha \in S$ such that $q \in \alpha$. Since α is a cut, $p \in \alpha$. So $p \in \lambda$.

3. Suppose $p \in \lambda$. So there is some $\alpha \in S$ such that $p \in \alpha$. Since α is a cut, there is some $q \in \alpha$ such that $p < q$. So $q \in \lambda$.

This proves the claim. Moreover, clearly $(\forall \alpha \in S)\alpha \subseteq \bigcup S = \lambda$, i.e. λ is an upper bound on S. So now suppose $\beta \in \mathbb{R}$ is also an upper bound, i.e. $(\forall \alpha \in S)\alpha \subseteq \beta$. For any $p \in \mathbb{Q}$, if $p \in \lambda$, then there is $\alpha \in S$ such that $p \in \alpha$, so that $p \in \beta$. Generalizing, $\lambda \subseteq \beta$. So λ is the *least* upper bound on S. $\square$